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Perinatal & Pediatric Pharmacology

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Objectives

- Understand how changes in physiology affect drug pharmacokinetics & pharmacodynamics in the fetus, infants, and children
- Know the functions of placenta & placental transporters
- Know the specific uses of corticosteroids, phenobarbital, indomethacin, alprostadil for neonates
- Know the FDA teratogenic risk categories
- Identify teratogens
- Know the cautions and contraindications in pregnancy and breastfeeding

A Child is NOT a Miniature Adult

Prior to the **PREA (Pediatric Research Equity Act) of 2003**

- Only **25%** of drugs used to treat children had actually been studied in children.

- **Three-quarters** of all medications marketed did *not* carry FDA approved labeling for use in neonates, infants, children and adolescents.

- Only **five** of the 80 drugs most frequently used in newborns and infants were labeled for pediatric use.

Slow Progress

Completion Rate and Reporting of Mandatory Pediatric Postmarketing Studies Under the US Pediatric Research Equity Act

Thomas J. Hwang, AB,[1,2] Liat Orenstein, MSc,[1,3] Aaron S. Kesselheim, MD, JD, MPH,[2] and Florence T. Bourgeois, MD, MPH[1,3,4]

JAMA Pediatr. 2019 Jan; 173(1): 68–74.

Conclusions and Relevance

Only 33.8% of mandatory pediatric postmarketing studies have been completed after a median follow-up of 6.8 years, and most drug labels do not include information important for pediatric use. To improve evidence-based prescribing of medicines to children, more timely completion of pediatric drug studies is needed.

But there is Progress

Off-Label Use of Drugs in Children

American Academy of Pediatrics: COMMITTEE ON DRUGS

Pediatrics March 2014, 133 (3) 563-567; DOI: https://doi.org/10.1542/peds.2013-4060

"The passage of the Best Pharmaceuticals for Children Act and the Pediatric Research Equity Act has collectively resulted in an improvement in rational prescribing for children, including more than 500 labeling changes. However, off-label drug use remains an important public health issue for infants, children, and adolescents, because an overwhelming number of drugs still have no information in the labeling for use in pediatrics. The purpose of off-label use is to benefit the individual patient. Practitioners use their professional judgment to determine these uses. As such, the term "off-label" does not imply an improper, illegal, contraindicated, or investigational use. Therapeutic decision-making must always rely on the best available evidence and the importance of the benefit for the individual patient."

Pediatric Dosage Forms

Elixirs: Drugs dissolved in alcoholic solutions

Suspensions: Drugs suspended in undissolved form; must be distributed throughout the vehicle by shaking; uneven distribution may cause inefficiency or toxicity (e.g., phenytoin suspension)

Remember pediatric dosages of medications are often weight based!

Pediatric Pharmacokinetics

Drug absorption

- **Blood flow** (IM, SC) *e.g., potential toxicity when using cardiac glycosides, aminoglycosides, anticonvulsant*

- **GI function:** GI function changes rapidly during the first few days after birth

e.g., pH, ↑ gastric emptying time, irregular or slow peristalsis, ↓ enzymes (e.g., α-amylase, lipase)

Pediatric Pharmacokinetics

Drug distribution

- Influenced by the changes of body composition

- **Water volume** (water-soluble drugs- aminoglycosides)
  - High water volume (75% vs. 60% in the adults), large extracellular space (40% vs. 20% in the adults)
- **Fat volume** (lipid-soluble drugs)
  - 1% in preterm infants vs. 15% in full-term infants
- **Binding to plasma proteins**
  - Generally ↓ in the neonates (↑ plasma free drug → ↑ drug toxicity)
  - **Sulfonamides compete with serum bilirubin for binding to albumin (cause kernicterus in the preterm neonates)**
  - ↑serum bilirubin may also displace a drug from albumin

Pediatric Pharmacokinetics

❑ Drug metabolism

- ↓ **liver enzyme activities**

e.g., Low activities of p450 enzyme (↑ chloramphenicol toxicity)

- ↑ **drug metabolism in toddlers** (12-36 months)

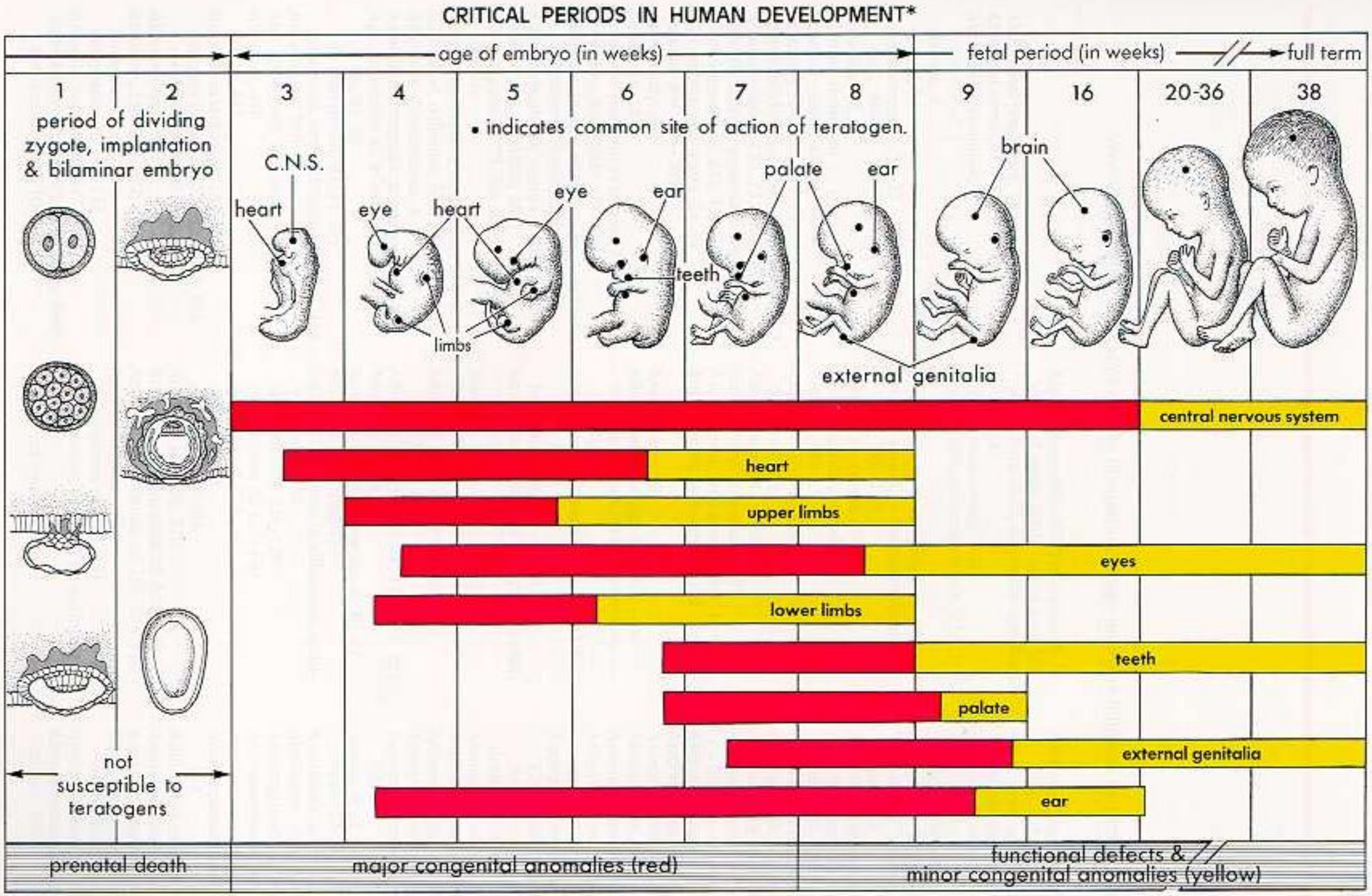
- ↓ **elimination half-life**

  e.g., maternal use of *phenobarbital* can induce early maturation of fetal hepatic enzyme, but may result in *increased drug metabolism in the neonates* (phenobarbital is a p450 inducer)

❑ Drug excretion

- ↓ **glomerular filtration rate**

e.g., ↑ renal toxicity: digoxin, penicillin, aminoglycosides



* Red indicates highly sensitive periods when teratogens may induce major anomalies.

Critical Factors Affecting Placental Drug Transfer and Drug Effects on the Fetus

- The physiochemical *properties* of the drug
- The *rate* at which the drug crosses the placenta
- The *amount* of drug reaching the fetus
- The *duration* of exposure to the drug
- *Distribution* characteristics in different fetal tissues
- The *stage* of placental and fetal development at the time of exposure to the drug
- The *effects* of drugs used in combination

Drug Transfer from the Mother to the Fetus

- **Lipid-solubility & drug ionization**
  - Lipophilic drugs tend to diffuse across placenta rapidly (e.g., thiopental).
  - Highly ionized drugs cross placenta slowly (e.g., succinylcholine)
  - *Thiopental and succinylcholine* are used for cesarean sections
  - *Thiopental* can cause sedation and apnea in newborn

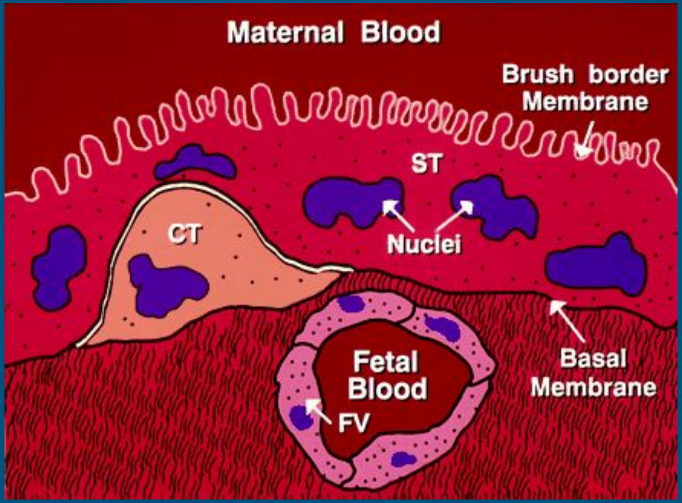
- **Molecular size**
  - Large molecules are harder to transfer
  - e.g., *heparin* is the choice for pregnant women (*warfarin is teratogenic*)

- **Protein-binding**
  - Drugs that have high affinity to maternal plasma proteins may be transferred slowly to the fetus (e.g., sulfonamides, local anesthetics)

- **Placental transporters**
  - Differential expression of transporters in the placenta affects the transfer of certain drugs/maternal antibodies

Placental Transporters

- Differentially expressed in a polarized manner in syncytiotrophoblast
- Facilitate the transfer of nutrients from the mother to the fetus
- Eliminate the metabolic waste from the fetus
- Prevent the transfer of xenobiotics to the fetus



ST: syncytiotrophoblast
CT: cytotrophoblast
FV: fetal blood vessel

https://www.openanesthesia.org/pharmacology_of_the_placenta/

DRUGS WHICH CROSS THE PLACENTA Potentially Dangerous

- Opiates (morphine, fentanyl if > 1 μg/kg)
- Benzodiazepines
- Ephedrine (increased metabolism)
- Local anesthetics
- Atropine
- β-blockers

DRUGS WHICH DO NOT CROSS THE PLACENTA

- All paralytics
- Glycopyrrolate
- Insulin
- Heparin

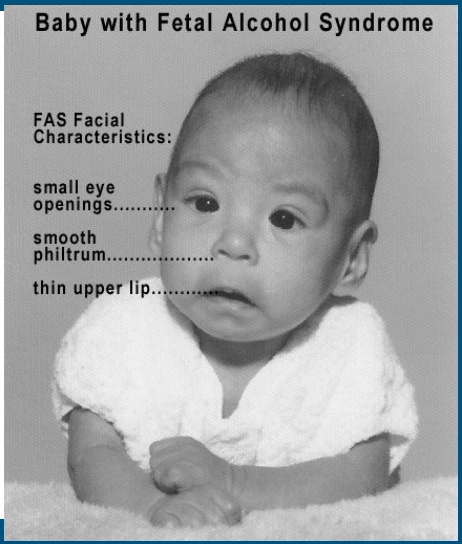
Safe

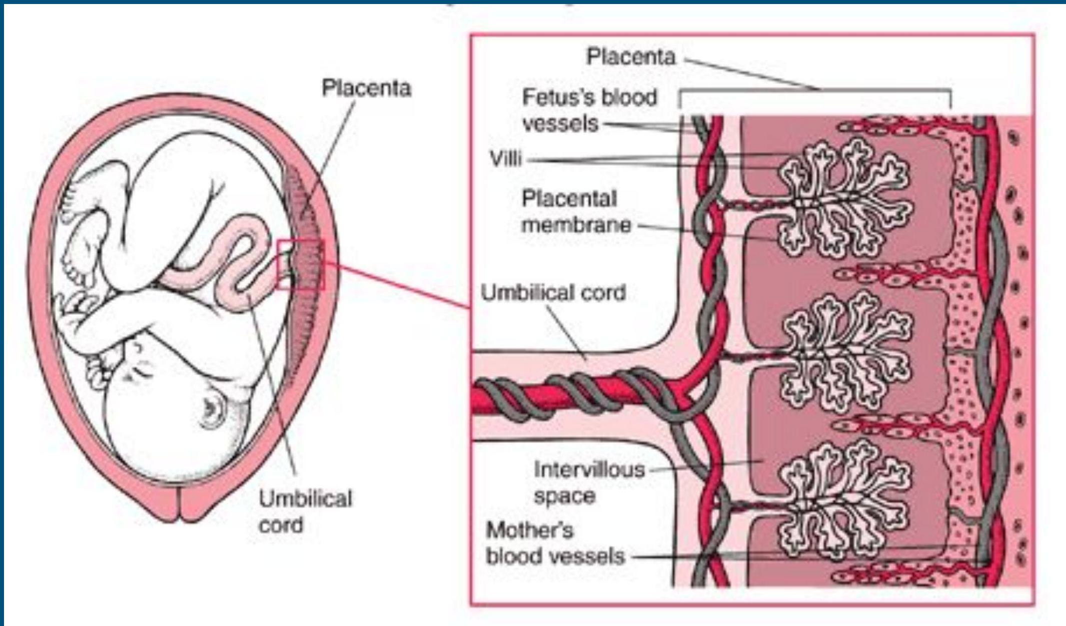
- Propofol
- Thiopental
- Ketamine
- Fentanyl at < 1 μg/kg
- Epidural opiates (fentanyl, sufentanil)

Placental and Fetal Drug Metabolism

❑ Placenta

- **A barrier**
- **A site of metabolism** (e.g. oxidation of pentobarbital)
- **May augment drug toxicity** (e.g., ethanol)





❑ Fetal liver

- **Some drugs may be partially metabolized before entering the fetal circulation**

Pharmacodynamics in the Fetus

Therapeutic drug actions

- **Betamethasone** (preterm birth) – *Stimulates* fetal lung maturation

- **Phenobarbital** (near term)- *Reduces* the incidence of jaundice (now less often used)

- **Antiarrhythmic drugs** (e.g., digoxin)- *Treat* fetal cardiac arrhythmias

- **Antiretroviral agents** (e.g., zidovudine)- *Decrease* vertical transmission of HIV

Pharmacodynamics in the Fetus

Predictable toxic drug actions in the fetus

- **Chronic use of opioids**
  Neonatal withdrawal syndrome

- **Angiotensin-converting enzyme inhibitors**: Severe renal damage in the fetus

- **Diethylstilbestrol (DES)**: Increased risk for adenocarcinoma of the vagina post- puberty

Teratogens & Teratogenesis

- **Teratogen:** A *substance* or *process* that
  - results in a characteristic set of malformations (selective for certain target organs)
  - exerts its effects at a particular stage of fetal development
  - shows a dose-dependent incidence

- **Teratogenesis:**

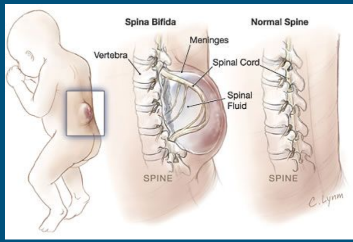
  The development of morphological malformation in the fetus/infant as a consequence of genetic abnormality or adverse environmental circumstances (e.g., exposed to thalidomide during 4-7 wks of gestation when the arms/legs develop)

FDA Teratogenic Risk Categories

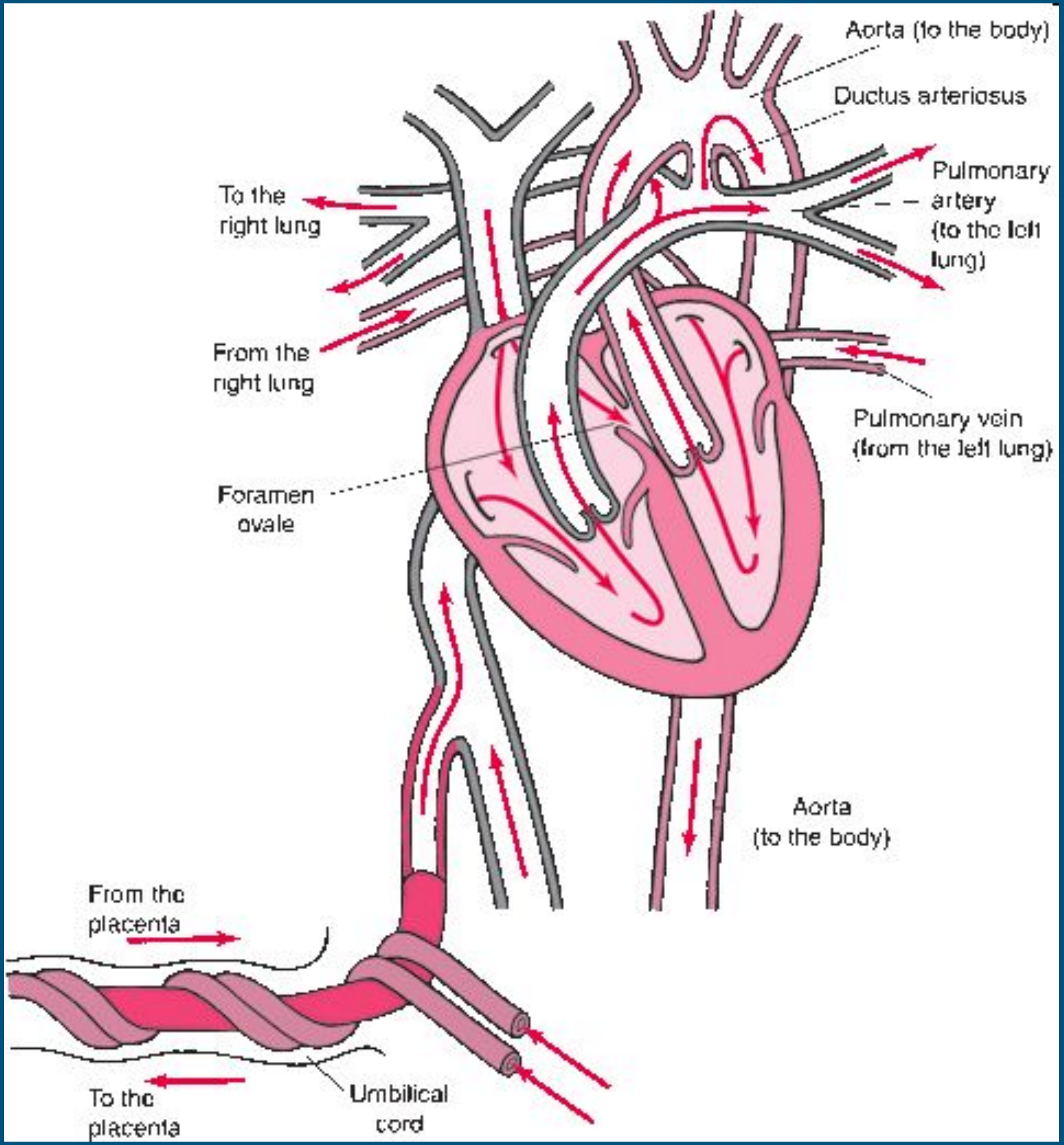
Category	Description
A	Controlled studies in women *fail to* demonstrate a risk to the fetus in the first trimester, and the possibility of fetal harm appears remote.
B	Either animal-reproduction studies have *not* demonstrated a fetal risk, but there are no controlled studies in pregnant women, or animal-reproduction studies have shown an adverse effect that was not confirmed in controlled studies in women in the first trimester.
C	Either studies in animals have revealed adverse effects on the fetus and there are *no* controlled studies in women or studies in women and animals are not available. Drugs should be given only if the potential benefit justifies the risk to the fetus.
D	There is *positive* evidence of human fetal risk, but the benefits from use in pregnant women may be *acceptable* despite the risk
X	Studies in animals or human beings have demonstrated fetal abnormalities, and the risk of the use of the drug in pregnant women clearly outweighs any possible benefit. The drug is *contraindicated* in women who are or may become pregnant.

Pharmacodynamics in the Fetus

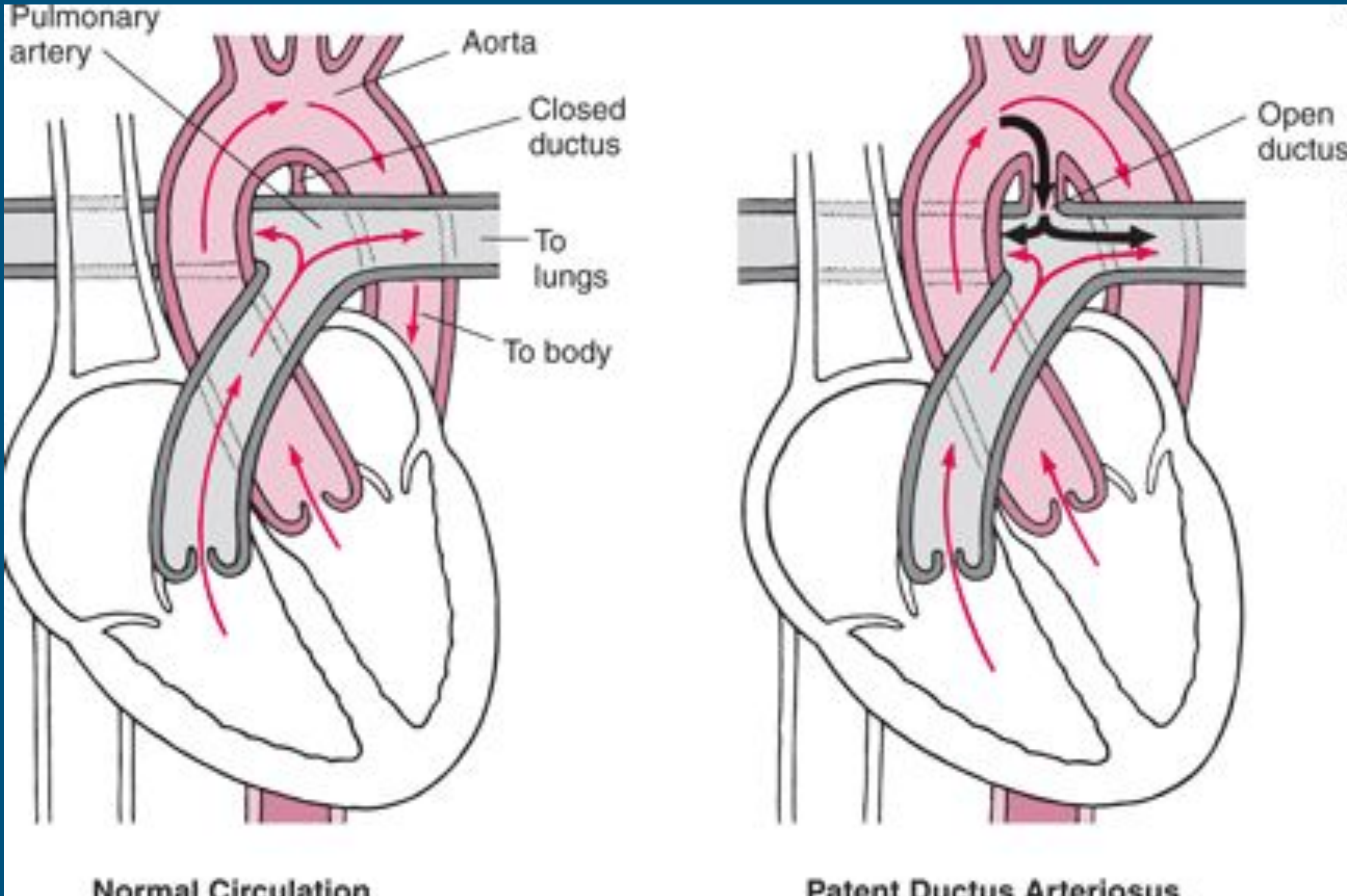
- **Teratogenic drug actions**
  - **Vit A analogs (retinoids)**
    e.g., isotretinoin (Accutane®), etretinate
    (removed from the US)



  - **High levels of homocysteine**
    - increases the risk of neural tube defects
    - Folate *deficiency* or genetic mutations (MTHFR) can cause hyperhomocysteinemia
  - Chronic consumption of **high doses of alcohol** during the $1^{st}/2^{nd}$ trimester (fetal alcohol syndrome)



Aorta (to the body)
Ductus arteriosus
Pulmonary artery (to the left lung)
To the right lung
From the right lung
Pulmonary vein (from the left lung)
Foramen ovale
Aorta (to the body)
From the placenta
To the placenta
Umbilical cord

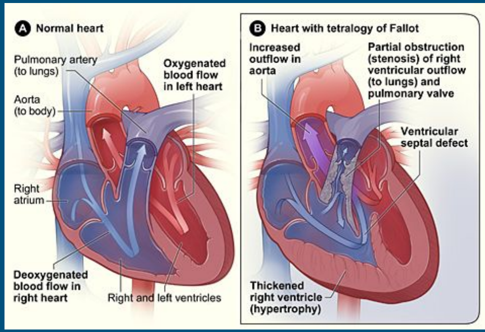


Pulmonary artery
Aorta
Closed ductus
To lungs
To body
Open ductus
Normal Circulation
Patent Ductus Arteriosus

Special Pharmacodynamic Features in the Neonate

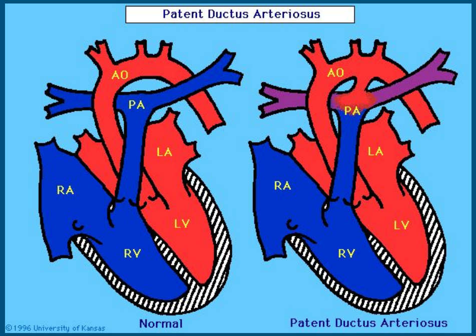
- **Prostaglandin E1 (alprostadil):**

  Prevents the closure of ductus arteriosus in infants with transposition of the great vessels or Tetralogy of Fallot (cyanosis-'blue baby')



- **Indomethacin**

  Induces rapid closure of a patent ductus arteriosus in preterm infants with respiratory distress syndrome



Medication Use During Lactation

- ❑ Medications for nursing women should be taken
  - 30-60 minutes after nursing
  - 3-4 hrs before the next feeding
- ❑ **Tetracycline**
  - in breast milk is 70% of the maternal plasma concentration (may cause permanent tooth staining in the infant)
- ❑ **Isoniazid**
  - may cause pyridoxine(B6) deficiency in the infant (B6 crucial for synthesis of serotonin, dopamine, epi, norepi
- ❑ **Sedatives and hypnotics** (withdrawal syndrome)
  - e.g., barbiturate, chloral hydrate, diazepam

Resources: Medications and Mother's Milk: Hale
Lactmed:https://www.ncbi.nlm.nih.gov/books/NBK501922/

Medication Use During Lactation

Cautions

- **Opioids** (heroin, methadone, morphine)
  - in breast milk is sufficient to prolong the state of neonatal narcotic dependence
- **Lithium**
  - achieves the same concentration in breast milk as in maternal serum

Breast-feeding is contraindicated ***after***

- **large doses of radioactive substances**
- **receiving cancer chemotherapy**
- **being treated with cytotoxic or immune modulating agents**

Drugs with Significant Adverse Effects on the Fetus

Drug	Trimester	Effect
ACE inhibitors	All	Renal damage
Aminopterin	1st	Multiple gross anomalies
Amphetamines	All	Suspected abnormal developmental patterns, decreased school performance
Androgens	2^{nd}, 3rd	Masculinization of female fetus
Tricyclic antidepressants	Third	Neonatal withdrawal symptoms have been reported in a few cases with clomipramine, desipramine, and imipramine
Barbiturates	All	Chronic use lead to neonatal dependence
Busulfan	All	Various congenital malformations; low birth weight
Carbamazepine	1st	Neural tube defects
Chlorpropamide	All	Prolonged symptomatic neonatal hypoglycemia
Clomipramine	3rd	Neonatal lethargy, hypotonia, cyanosis, hypothermia
Cocaine	All	Increased risk of spontaneous abortion, abruption placenta, and premature labor; neonatal cerebral infarction, abnormal

Click on the hyperlinks for more

Drugs with Significant Adverse Effects on the Fetus

Drug	Trimester	Effect
Cyclophosphamide	**1st**	**Various congenital malformations**
Cytarabine	**1st, 2nd**	**Various congenital malformations**
Diazepam	**All**	**Chronic use may lead to neonatal dependence**
Diethylstilbestrol	**All**	**Vaginal adenosis, clear cell vaginal adenocarcinoma**
Ethanol	**All**	**Risk of fetal alcohol syndrome and alcohol-related neurodevelopmental defects**
Etretinate	**All**	**High risk of multiple congenital malformations**
Heroin	**All**	**Chronic use leads to neonatal dependence**
Iodide	**All**	**Congenital goiter, hypothyroidism**
Isotretinoin	**All**	**Extremely high risk of CNS, face, ear, and other malformations**
Lithium	**1st**	**Ebstein's anomaly**
Methadone	**All**	**Chronic use leads to neonatal dependence**
Methotrexate	**1st**	**Multiple congenital malformations**

Click on the hyperlinks for more

Drugs with Significant Adverse Effects on the Fetus

Drug	Trimester	Effect
Methylthiouracil	All	Hypothyroidism
Metronidazole	First	May be mutagenic (from animal studies)
Organic solvents	First	Multiple malformations
Misoprostol	1st	Moebius sequence
Penicillamine	1st	Cutis laxa, other congenital malformations
Phencyclidine	All	Abnormal neurologic examination, poor suck reflex and feeding
Phenytoin	All	Fetal hydantoin syndrome
Propylthiouracil	All	Congenital goiter
Streptomycin	All	Eighth nerve toxicity
Smoking	All	Intrauterine growth retardation; prematurity; sudden infant death syndrome; perinatal complications

Click on the hyperlinks for more

Drugs with Significant Adverse Effects on the Fetus

Drug	Trimester	Effect
Tamoxifen	All	Increased risk of spontaneous abortion or fetal damage
Tetracycline	All	Discoloration and defects of teeth and altered bone growth
Thalidomide	1st	Phocomelia (shortened or absent long bones of the limbs) and many internal malformations
Trimethadione	All	Multiple congenital anomalies
Valproic acid	All	Neural tube defects, cardiac and limb malformations
Warfarin	1st	Hypoplastic nasal bridge, chondrodysplasia
	2nd	CNS malformations
	3rd	Risk of bleeding. Discontinue use 1 month before delivery

Click on the hyperlinks for more